

6 Results

Unless stated otherwise, we report query-macro percentages on the primary populations, percentage-point (pp) differences, and nDCG on the 0–1 scale.

6.1 Attribute-Order Sensitivity (RQ1)

Across the seven catalog-wide schedules, VI@20 ranges from 11.93–19.58% on WANDS and 3.66–6.91% on ESCI. These estimates describe crossing within the tested family, not a probability of omission under future orders.

With every competitor fixed at source order, changing only the target yields VI@20 of 10.83–19.45% on WANDS and 2.42–5.88% on ESCI. The interventions answer complementary questions; their difference does not decompose target and competitor effects. Sensitivity remains when every tested target input fits completely: fully-fitting VI@20 ranges from 7.16–14.55% and 2.55–5.88%, respectively (Figure 3). GTE’s full and fitting populations coincide. MiniLM and BGE fitting subsets differ from the full populations and from one another, so these comparisons do not isolate a causal truncation or architecture effect.

On the frozen 128-query WANDS supplement, expanding the harmonized family from seven to 32 schedules increases target-only VI@20 by 7.18 pp (95% CI [5.73, 8.89]) for MiniLM and 6.09 pp (95% CI [4.56, 7.81]) for BGE, with the full execution comparison and common fitting supports in the supplement.

6.2 Aggregate Effectiveness and Product Membership (RQ2)

Gains and losses can be substantially larger than their net Recall change. Reversing WANDS attributes with BGE newly includes 2.357% and newly omits 2.263% of highest-label support. The full-precision sum rounds to 4.621%, while Recall changes by only +0.094 pp (95% CI [−0.409, 0.563]). Each share is divided by $|H_q|$ within a query before macro averaging.

Exact replacement also occurs within individual queries. For this reversal, 73 of 308 WANDS/BGE queries (23.70%) have equal positive gain and loss counts. Across the six contrasts, counts range from 67 to 78 (21.75–25.32% per contrast), not a union of distinct queries. Figure 4 uses reversal as the same definition-based editorial contrast in every setting, selected after historical outcomes; all original schedules remain in companion outputs. In the same catalog-wide reversal, an illustrative hash-selected query, “anti fatigue mat,” exchanges five Exact products in each direction while Recall@20 remains exactly 20/307. The case companion records every changed ID, name, label, rank and order.

Graded effectiveness is complementary. Conditional on changed Top-20 relevant membership, median absolute nDCG@20 differences range from 0.0226–0.0399 on WANDS and 0.0388–0.0482 on ESCI. Signed means over all eligible queries range from −0.01101 to +0.00151 and −0.00393 to +0.00092, respectively. These summarize different populations and quantities. The companion retains each named contrast, its unadjusted interval, condensed nDCG and judged coverage; intervals containing zero do not establish equality.

6.3 Effectiveness and Consistent Inclusion under Controls (RQ3)

Canonical, set-based, fixed-view and lexical controls remove incoming-order dependence by construction. Their zero VI is structural; effectiveness and which products are persistently included remain empirical questions. Table 1 retains all three canonical rules without choosing a test winner. Raw dense and raw hybrid effectiveness average seven schedules; source-order results remain separately labelled in the companion.

Simple controls have setting-dependent effectiveness. Set-Mean’s Recall@100 exceeds the raw dense seven-schedule mean on WANDS/MiniLM and is lower in the other three MiniLM/BGE settings. Raw hybrid minus raw dense changes WANDS/BGE mean Recall@100 by +4.152 pp (95% CI [2.551, 5.819]); full paired comparisons for all settings accompany the table. These observations do not establish a general effectiveness–instability trade-off.

For a convention-based canonical example, WANDS/BGE ascending canonical hybrid changes nDCG@20 by +0.0691 (95% CI [0.0554, 0.0832]) versus BM25 and +0.0259 (95% CI [0.0142, 0.0378]) versus its corresponding ascending dense branch. Against the raw-hybrid seven-schedule mean, its Recall@20 changes by −0.624 pp (95% CI [−1.370, −0.084]). This counterexample remains visible even though the main table reports R@100. Each interval is unadjusted; neither an interval crossing zero nor structural invariance establishes universal preservation.

The fixed order can place a previously crossing product on either side of the cutoff (Table 2). In the dense example, 7.57% of all support moves from crossing to inclusion and 12.01% to omission. The control also omits 0.23% that was persistently included and includes 0.43% that was persistently omitted. These are unconditional accounting shares, not newly caused losses or failure probabilities. The two small stable-state cells alone do not recover total Recall change because crossing products also contribute. The canonical order may lie outside the seven raw orders.

Cost provides a separate comparison. Canonicalization retains one dense vector; centroid construction encodes M views but retains and scores one vector, whereas multi-vector retrieval stores and scores M vectors. Hybrid adds sparse storage and branch scoring. Existing cost measurements use warm-cache CPU exact retrieval and exclude query encoding; they are not ANN deployment latency. Effectiveness, persistent inclusion/omission, identities and cost therefore accompany VI.

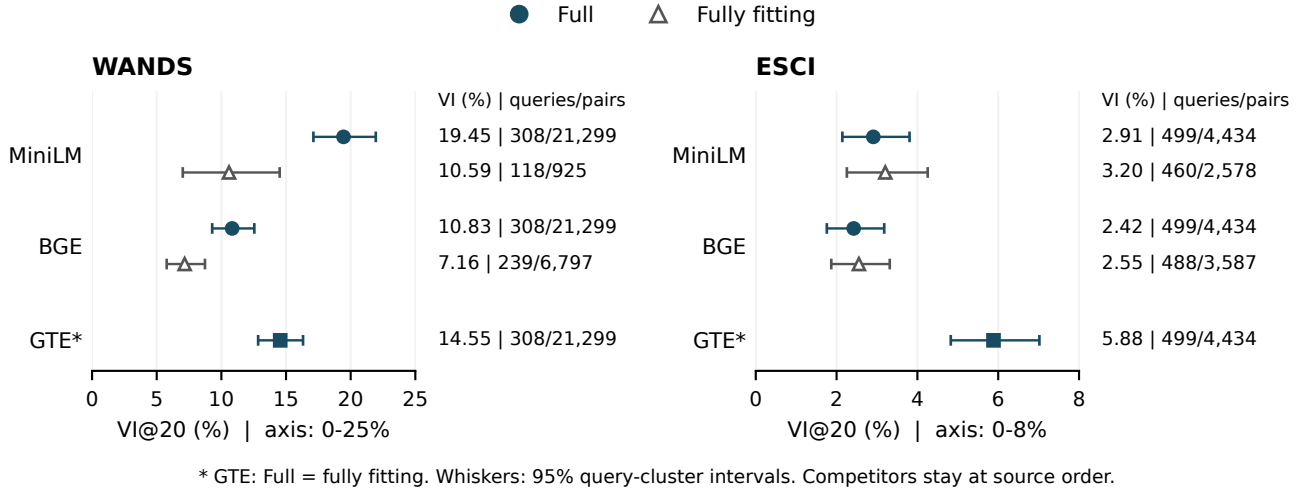


Figure 3: Target-only VI@20 with 95% query-cluster intervals and query/pair support. GTE shows one estimate for identical full and fitting populations. Panel axis ranges differ explicitly; competitors remain fixed.

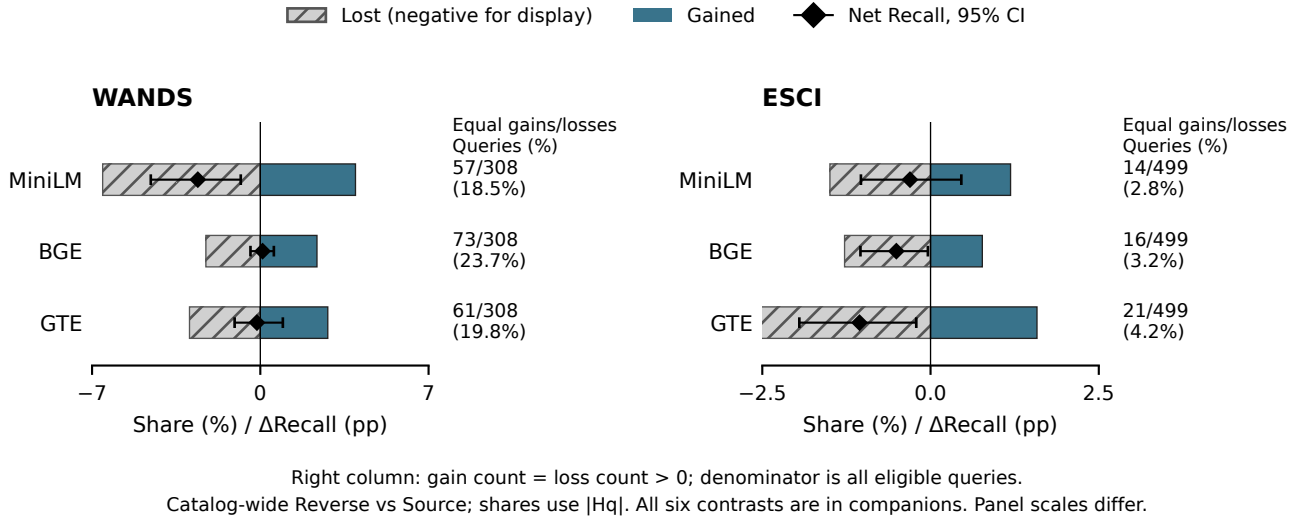


Figure 4: Catalog-wide Reverse versus source at $K = 20$. Bars show query-macro gained/lost shares of H_q ; losses are nonnegative quantities drawn left only for display. Diamonds show net Recall differences with 95% paired intervals. Counts require equal positive integer gains and losses and use all eligible queries as denominator.

Table 1: Common-support controls. Recall (R) and VI are percentages; nDCG uses 0–1. R@100/VI@100 assess candidate inclusion, whereas nDCG@20 assesses ranking near the top. All fixed controls have structural VI=0. The companion includes R@20, paired intervals, source-order references, GTE and every view budget.

Method	MiniLM			BGE		
	R@100	VI@100	nDCG@20	R@100	VI@100	nDCG@20
WANDS						
Raw dense (7 mean)	56.61	21.89	0.6323	63.62	12.30	0.7008
Canonical: ascending	56.05	0.00	0.6264	63.24	0.00	0.6981
Canonical: descending	56.16	0.00	0.6341	63.61	0.00	0.7023
Canonical: type priority	57.19	0.00	0.6312	63.56	0.00	0.7030
BM25	63.23	0.00	0.6548	63.23	0.00	0.6548
Raw hybrid (7 mean)	65.55	11.08	0.6983	67.77	7.05	0.7254
Canonical hybrid: ascending	65.61	0.00	0.6960	68.15	0.00	0.7239
Canonical hybrid: descending	65.29	0.00	0.6979	67.93	0.00	0.7272
Canonical hybrid: type priority	65.71	0.00	0.6988	67.83	0.00	0.7262
Set-Mean	58.81	0.00	0.6366	61.82	0.00	0.6841
ESCI						
Raw dense (7 mean)	86.19	1.41	0.6019	91.04	0.72	0.6623
Canonical: ascending	86.41	0.00	0.6015	91.05	0.00	0.6632
Canonical: descending	86.38	0.00	0.6008	91.01	0.00	0.6612
Canonical: type priority	86.36	0.00	0.6013	91.09	0.00	0.6629
BM25	82.83	0.00	0.6151	82.83	0.00	0.6151
Raw hybrid (7 mean)	89.57	0.56	0.6404	91.61	0.74	0.6690
Canonical hybrid: ascending	89.51	0.00	0.6392	91.55	0.00	0.6710
Canonical hybrid: descending	89.66	0.00	0.6405	91.55	0.00	0.6690
Canonical hybrid: type priority	89.54	0.00	0.6392	91.57	0.00	0.6709
Set-Mean	84.50	0.00	0.5887	88.95	0.00	0.6371

Table 2: One transition illustration: WANDS/MiniLM ascending at $K = 20$. Cells are query-macro percentages of all highest-label support. Dense and hybrid use their own raw families; each six-cell block sums to 100% before rounding.

Raw reference state	Dense		Hybrid	
	Included	Omitted	Included	Omitted
Persistent inclusion	20.72	0.23	29.27	0.21
Crossing	7.57	12.01	6.77	6.93
Persistent omission	0.43	59.04	0.42	56.40